

2 Weighted Voting

2.3 Shapley-Shubik Power Index

Definition 2.3.1 A sequential coalition is an ordered coalition, that is, it keeps track of the order in which each player joins the coalition.

We will use the notation

$$\langle P_1, P_2, P_3 \rangle$$

to denote sequential coalitions.

Definition 2.3.2 A **pivotal player** is the player in the coalition which changes the coalition from a losing coalition to a winning one.

Definition 2.3.3 (Shapley-Shubik power index) To compute the Shapley-Shubik power index:

- List all sequential coalitions
- Determine all pivotal players
- Count up how many times each player was pivotal
- convert these into fractions by dividing each count by the total number of sequential coalitions

Example 2.3.4 Calculate the Shapley-Shubik Power Index for the following weighted voting system

$$[8 : 6, 3, 2].$$

Power index is:

$$P_1 : 2/3, P_2 : 1/6, P_3 : 1/6.$$

Example 2.3.5 Compute the Shapley-Shubik Power Index for the weighted voting system:

$$[36 : 20, 17, 15]$$

$$P_1 = 1/2, P_2 = 1/2, P_3 = 0.$$

Example 2.3.6 Compute the Shapley-Shubik Power Index for the weighted voting system:

$$[22 : 23, 15, 7].$$

Notice that P_1 is **not** a dictator!

Example 2.3.7 Small review: Compute the Banzhaf Power Index for the weighted voting system:

$$[39 : 23, 17, 15].$$

Are there any dummies in this system?

Power index:

$$P_1 : 1/2, P_2 : 1/2, P_3 : 0.$$

Example 2.3.8 Consider the weighted voting system given by

$$[q : 7, 3, 1].$$

Which values of q result in a dictator? (list all possible values)

Example 2.3.9 In the Shapley-Shubik method, is it possible for a dummy to be a pivotal player? Why or why not?

No, if they were, they would be a critical player in a winning coalition.